An-Sheng (Anson) LU

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EDUCATION

University of California, Riverside, CA, USA | *Expected Spring 2029* **2025 – Present**

Major in Electrical Engineering at Department of Electrical and Computer Engineering | Cumulative GPA: 3.74.
Relevant Coursework: Multivariable Calculus, Physics, Data Structures & Algorithms (C++).
Scholarship: UCR Achievement Scholarship (2025–29).

Victoria Academy, Yunlin, Taiwan **2022 – 2025**

IB Bilingual Diploma, July 6, 2025 | GPA: 3.94 (IBDP) / 3.86 (Cumulative).
- International Baccalaureate Diploma Programme (IBDP) HL subjects: Mathematics AA (7), Physics (6), Chemistry (6).

Graduation Awards: Chairman's/Mayor's Award, Dr. Cho Rui-Chiao Scholarship, Excellence Award in Mathematics.

EXPERIENCES

Intern, Electrical Subsystems, Highlander Racing (FSAE), University of California, Riverside **Apr. 2026 – Present**

- Contributed to the design of an industrial automation PCB in Altium Designer, from schematic capture of analog sensing stages to routing power stages.

Research Assistant, Robotics and Medical Systems (RaMS) Lab, University of California, Riverside **Oct. 2025 – Present**

- Independently developed a CV/DL pipeline using AISFormer in collaboration with the UCR Entomology Dept., automating biomass estimation and growth stage classification for mealworms; presented findings in biweekly meetings with PI.
- Architected automated workflows featuring normal distribution filtering and sensor calibration to translate image data into high-accuracy phenotypic measurements for life-cycle analysis.

Inventor, Taiwan Intellectual Property Office, Taiwan **June 2024 – Aug. 2025**

- Patent: *Baton Device with Directional Indication* (Invention Patent No. I894025). Taiwan Patent Search System.
- Independently designed a signaling device with custom hardware logic using an ESP32 and MPU-6050 (IMU) to process real-time indication; managed conceptual design and circuit prototyping phases through to formal patent approval.

Intern, Robotics and Mechatronics Center, German Aerospace Center (DLR), Oberpfaffenhofen, Germany **Aug. 1 – 30, 2024**

- Integrated a custom robotic system using an OpenManipulator-X arm; collaborated with advisors on system design and implemented 3D-printed structural interfaces and motion control.
- Modeled mass-spring-damper systems in MATLAB/Simulink to analyze resonance and power dissipation; designed a critically damped feedback control system to optimize stability.
- Designed and assembled a 3x3x3 LED matrix using an Arduino Micro; implemented a 4-mode firmware system featuring real-time pace control via ADC and a variable resistor.

Robotics Club Member, National Yunlin Univ. of Sci. & Tech. (NYUST), Yunlin, Taiwan **Sep. 2023 – June 2024**

- Collaborated with club members to design an omnidirectional mobile robot with a three-Mecanum wheel chassis; executed sensor fusion using Ultrasonic and Color Detection sensors for obstacle avoidance and environmental interaction via LabVIEW.
- Implemented basic ROS interfacing for system communication.

HONORS & AWARDS

- 1st Place Winner, Robotics Competition, Yunlin, Taiwan, Dec. 25, 2023.
- 1st Place Winner, Industrial Robot Competition, Kaohsiung, Taiwan, Mar. 9, 2024.
- Final Shortlisted Prize, Electrical Engineering Category, 2024 All Ring Competition, Tainan, Taiwan, Oct. 24, 2024
- Victoria Academy Scholarships (5x recipient, 2022–2024).

INDEPENDENT ACADEMIC STUDIES

- **Determination of gravity vector** (Oct. 2024): Identified gravity vector in body-fixed frame using tri-axial accelerometer.
- **Resonance in a shaking device** (Mar. 2025): Analyzed power dissipation changes with spring constant variations.

SKILLS

- **Language:** Mandarin (Native), English (Fluent), Taiwanese/Min Nan Chinese (Conv.), German (Basic)
- **Programming:** C/C++, Python (PyTorch, OpenCV), LabVIEW, MATLAB/Simulink, ROS (Basic).
- **Software & Hardware Skills:** Tinkercad, Creo, Arduino (IDE), Visual Studio/VS Code, Altium Designer, 3D Printing.
- **Interest:** Basketball (Clippers), Formula 1 (Red Bull Racing), Strategic Games (Chess/Xiangqi), Robotics, Electronics.